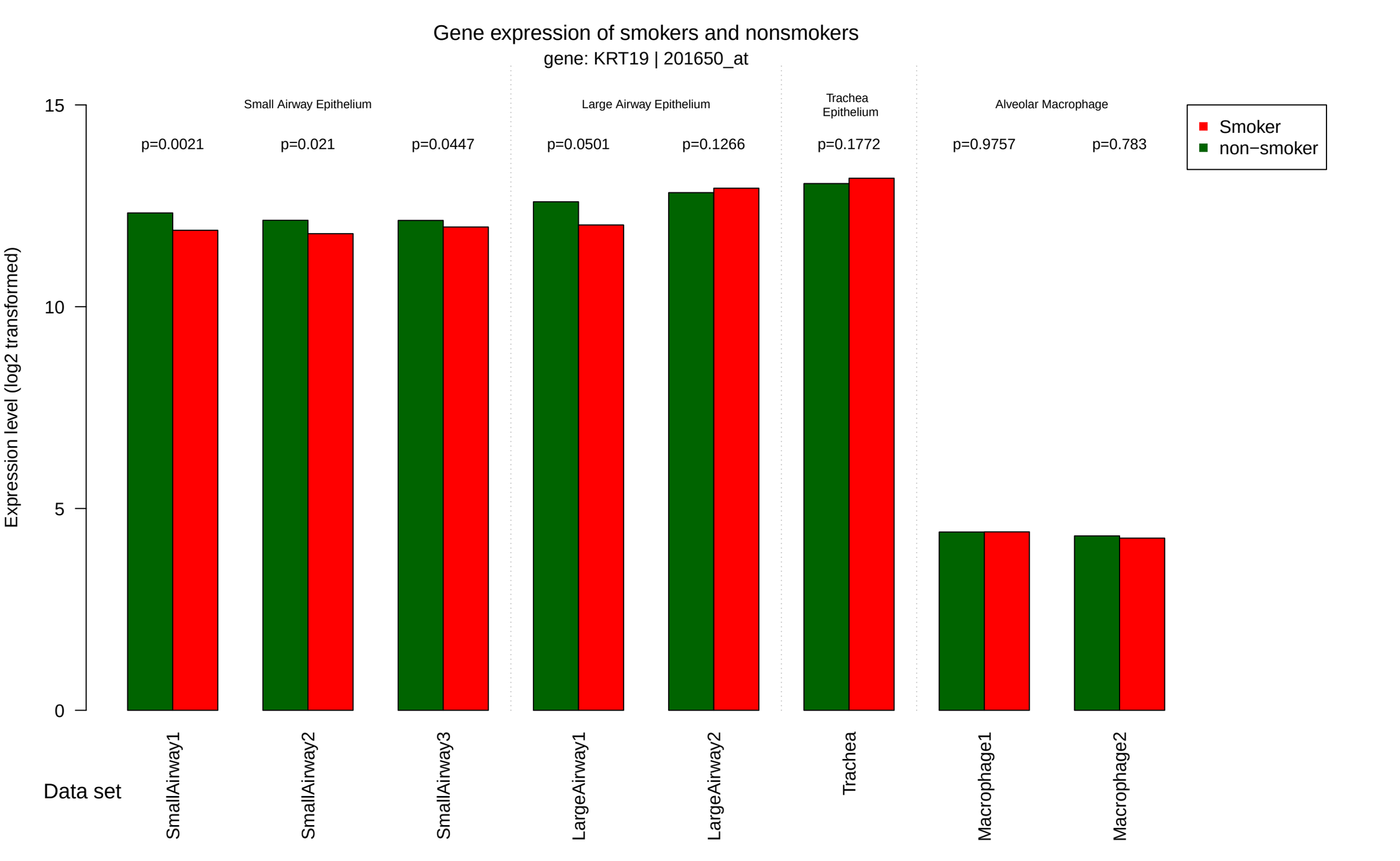
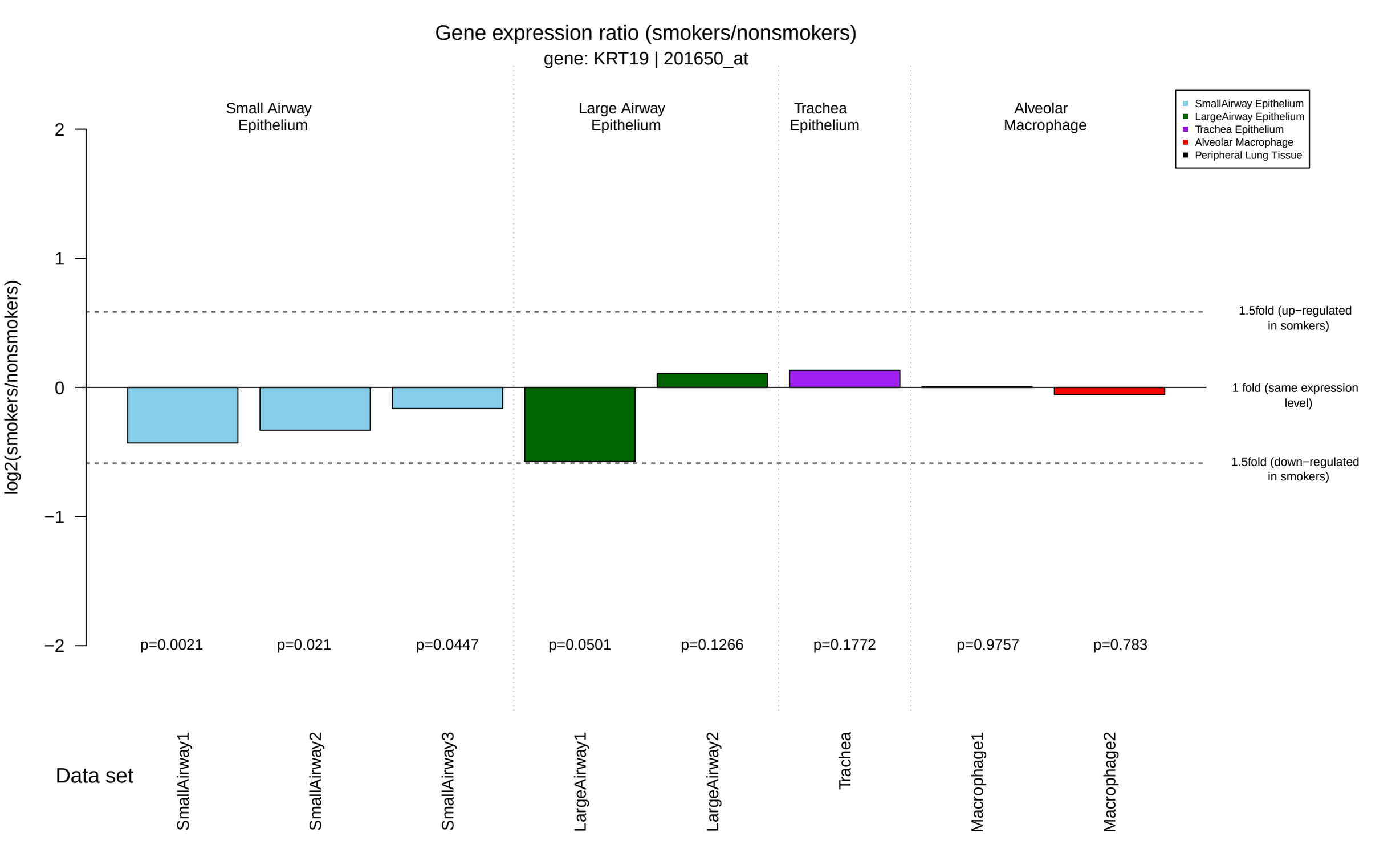
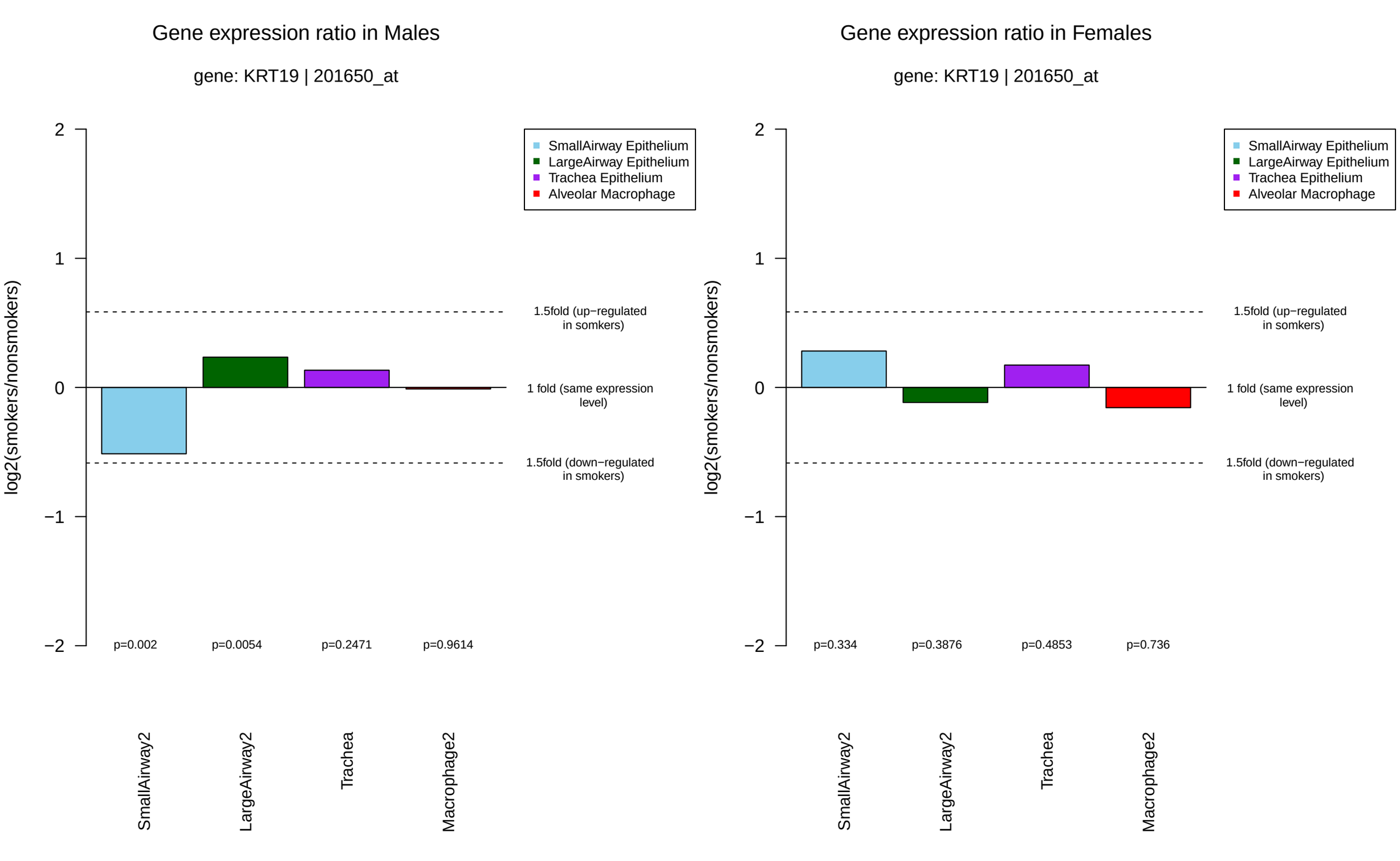
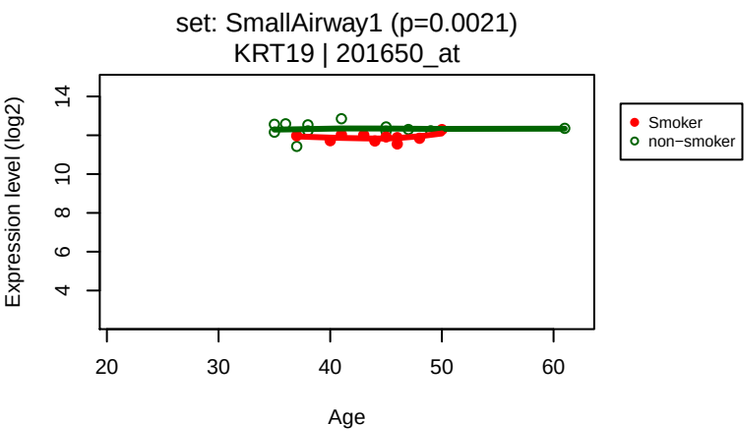
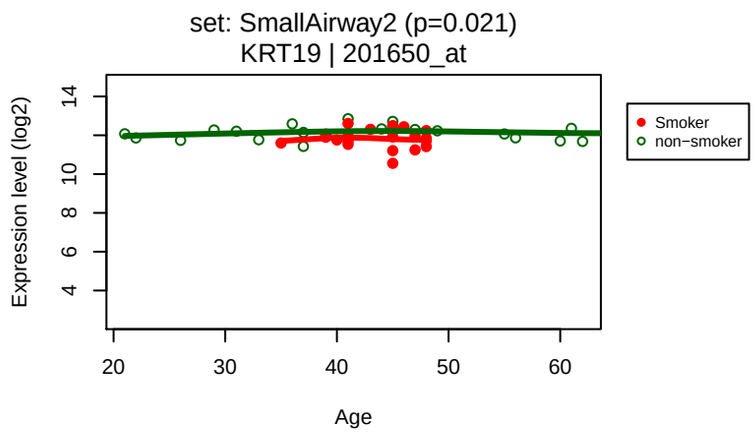
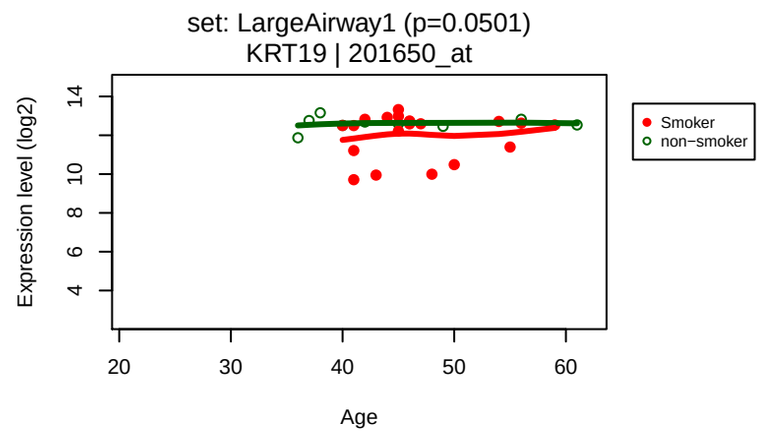

Gene expression of smokers and nonsmokers
gene: KRT19 | 201650_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0021
p=0.021
p=0.0447
p=0.0501
p=0.1266
p=0.1772
p=0.9757
p=0.783
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

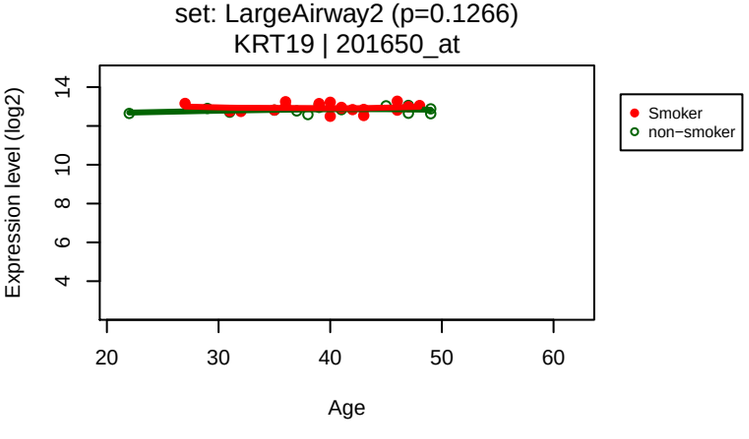

Gene expression ratio (smokers/nonsmokers)
gene: KRT19 | 201650_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0021
p=0.021
p=0.0447
p=0.0501
p=0.1266
p=0.1772
p=0.9757
p=0.783
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

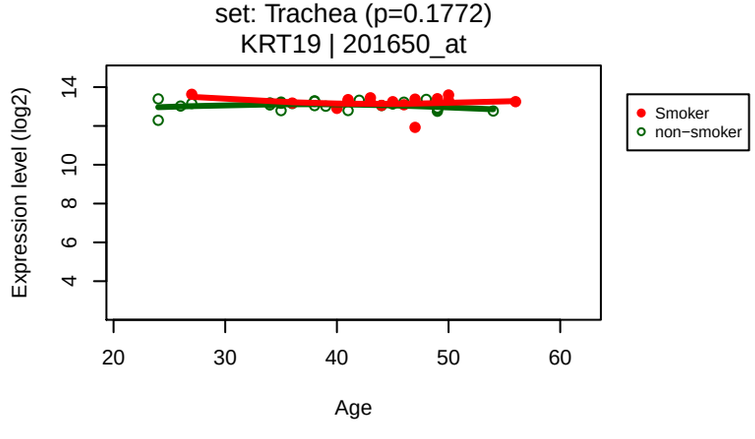

Gene expression ratio in Males
gene: KRT19 | 201650_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.002
p=0.0054
p=0.2471
p=0.9614
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KRT19 | 201650_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.334
p=0.3876
p=0.4853
p=0.736
SmallAirway2
LargeAirway2
Trachea
Macrophage2

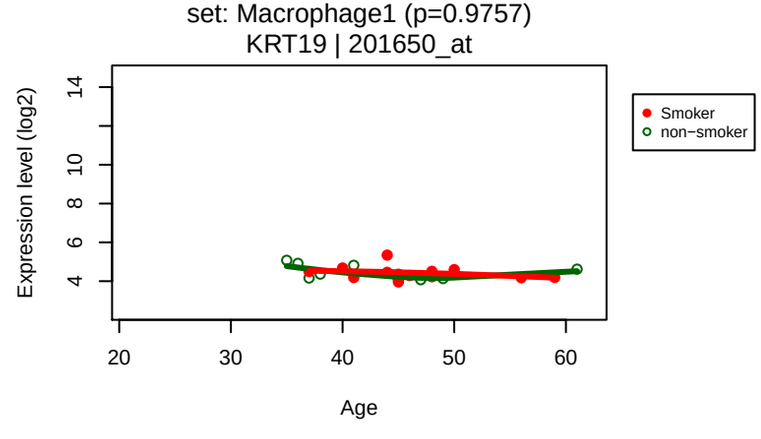

set: SmallAirway1 (p=0.0021)
KRT19 | 201650_at
Expression level (log2)
Age
Smoker
non−smoker

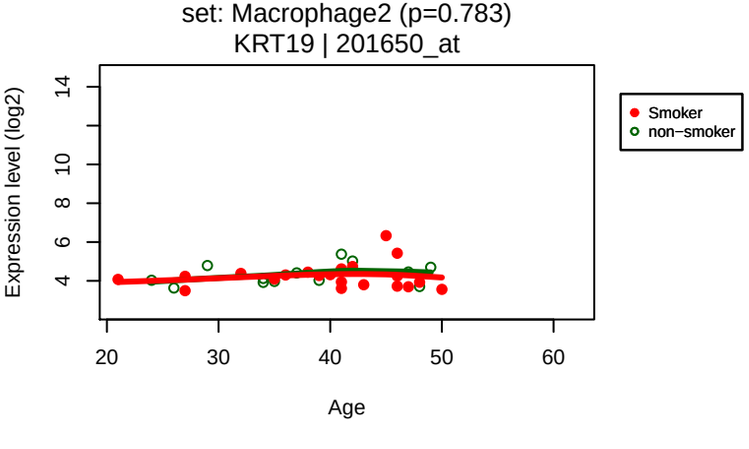

set: SmallAirway2 (p=0.021)
KRT19 | 201650_at
Expression level (log2)
Age
Smoker
non−smoker

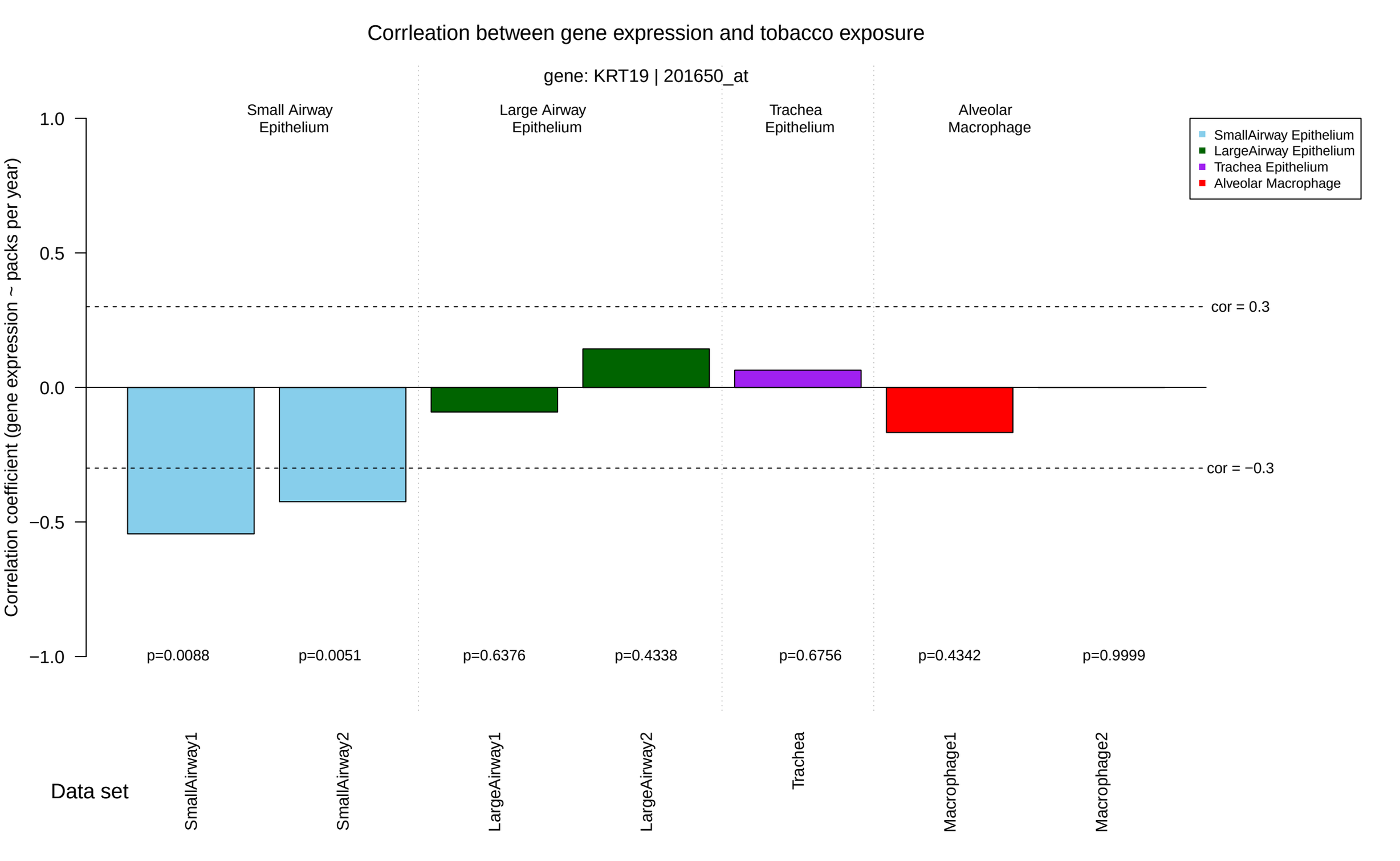

set: LargeAirway1 (p=0.0501)
KRT19 | 201650_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1266)
KRT19 | 201650_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1772)
KRT19 | 201650_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9757)
KRT19 | 201650_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.783)
KRT19 | 201650_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KRT19 | 201650_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0088
p=0.0051
p=0.6376
p=0.4338
p=0.6756
p=0.4342
p=0.9999
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2